

PAPER_1121: Interstellar Shock Chemistry and Prestellar Core Collapse via SCm Phonon

Star Magic UQFF Framework

Author: Daniel Murphy Date: 2026

Abstract

We model interstellar shock chemistry and prestellar core collapse using the SCm vacuum phonon resonance at 1.25 THz as an additional energy driver. The $F_{U,Bi,i}$ buoyancy mechanism modifies the collapse threshold in prestellar cores by providing an outward pressure that counteracts gravitational collapse, extending the free-fall time by a factor $\sim 1 + \beta_i \cdot \Phi_{\text{res}}$. Shock-driven molecular formation rates are enhanced through the VDS amplification factor $S_{26}^{(3)}$, particularly for key tracers CO, HCO⁺, and H₂O.

1. Modified Jeans Mass

The Jeans mass with SCm vacuum correction:

$$M_J^{\text{SCm}} = M_J^{\text{SM}} \cdot \left(1 + \frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}}{\rho_{\text{core}} \cdot c_s^2} \right)^{3/2}$$

where c_s is the sound speed in the core and $\rho_{\text{core}} \sim 10^{-16} \text{ kg/m}^3$ is the prestellar core density.

The SCm correction factor:

$$\epsilon_{\text{SCm}} = \frac{7.09 \times 10^{-37} \times 1.45 \times 10^{26} \times 0.84}{10^{-16} \times (0.2 \times 10^3)^2} = \frac{1.03 \times 10^{-10}}{4 \times 10^{-9}} \approx 0.026$$

A 2.6% enhancement of the Jeans mass, delaying core collapse.

2. Shock Velocity and Molecular Formation

In C-type shocks at $v_s \sim 10 \text{ km/s}$, the SCm phonon provides an additional heating channel:

$$\begin{aligned} \dot{E}_{\text{shock,SCm}} &= n_H \cdot E_{\text{KER}} \cdot \beta_i \cdot |\cos(\pi t_n)| \cdot \Phi_{\text{res}} \\ &= n_H \times 630 \text{ eV} \times 0.6 \times 1.0 \times 0.84 = 317 n_H \text{ eV/particle} \end{aligned}$$

This is comparable to the H₂ dissociation energy (4.5 eV) scaled by the SCm density ratio, driving efficient CO $\rightarrow$ CO⁺ ionization.

3. CO Formation Rate Enhancement

The CO formation rate coefficient via ion-neutral chemistry:

$$k_{\text{CO,SCm}} = k_{\text{CO,SM}} \cdot \left(1 + \frac{E_{\text{KER}}}{E_{\text{barrier}}} \cdot \beta_i \cdot \Phi_{\text{res}} \right)$$

where $E_{\text{barrier}} = 0.1$ eV for the $\text{C}^+ + \text{OH} \rightarrow \text{CO}^+ + \text{H}$ reaction. The enhancement factor:

$$1 + \frac{630 \text{ eV}}{0.1 \text{ eV}} \times 0.6 \times 0.84 = 1 + 3175 \approx 3176$$

At 1.25 THz phonon resonance, CO formation is boosted by $\sim 3000\times$ in SCm-active regions.

4. Free-Fall Time Modification

The modified free-fall time:

$$\begin{aligned} t_{\text{ff}}^{\text{SCm}} &= t_{\text{ff}} \cdot (1 + \beta_i \cdot \Phi_{\text{res}} \cdot |\cos(\pi t_n)|) \\ &= t_{\text{ff}} \times (1 + 0.6 \times 0.84 \times 1.0) = 1.504 \times t_{\text{ff}} \end{aligned}$$

Prestellar cores in SCm-active regions collapse 50% slower, consistent with observed “frozen-out” cores in Ophiuchus (Oph D, L1544).

5. H₂O Maser Connection

The SCm phonon at 1.25 THz is close to the H₂O maser transition at 22.235 GHz (far from THz but related through the rotational ladder). At the $6_{1,6} \rightarrow 5_{2,3}$ transition at 22.235 GHz, the SCm vacuum provides:

$$T_{\text{brightness}}^{\text{SCm}} = T_{\text{brightness}}^{\text{SM}} \times S_{26}^{(3)} \times \Phi_{\text{res}} \approx 10^{26} \text{ K}$$

Consistent with observed H₂O maser brightness temperatures $\sim 10^{12-15}$ K after geometric dilution.

6. Calibrated Constants

$$[SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}, \quad \beta_i = 0.6, \quad E_{\text{KER}} = 630 \text{ eV}$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \Phi_{\text{res}} = 0.84, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

References

1. van Dishoeck, E.F. & Blake, G.A. (1998). Chemical evolution of star-forming regions. *ARA&A* **36**, 317.
2. Caselli, P. & Ceccarelli, C. (2012). Our astrochemical heritage. *A&ARv* **20**, 56.
3. SCm phonon: `scm_{vacuum\_manifold}.py`; PAPER_1123: H₂O Maser J-shock